25. When a cluster of genes show linkage behaviour they (AIPMT 2003)
a) do not show a chromosome map
b) show recombination during meiosis
c) **do not show independent assortment**
d) induce cell division

26. Genetic map is one that (AIPMT 2003)
a) **Establish sites of the genes on a chromosome**
b) Establishes the various stages in gene evolution
c) Shows the stages during the cell division
d) Shows the distribution of various species in a region

27. After a mutation at a genetic locus of the character of an organism changes due to the change in (AIPMT 2004)
a) DNA replication
b) Protein synthesis pattern
c) RNA transcription pattern
d) **Protein structure**

28. In a hexaploidy wheat, the haploid (n) and basic (x) numbers of chromosomes are (AIPMT 2007)
a) **n =21 and x =7** b) n =7 and x =21
c) n =21 and x =21 d) n =21 and x =14

29. Point mutation involves (AIPMT 2009)
a) Deletion b) Insertion
c) **Change in single base pair**
d) duplication

30. Which one of the following is a wrong statement regarding mutations**?** (AIPMT 2012)
a) UV and Gamma rays are mutagens
b) **Change in a single base pair of DNA does not cause mutation**
c) Deletion and insertion of base pairs cause frame shift mutations.
d) Cancer cells commonly show chromosomal aberrations.

31. Which of the following statement is not true of two genes that show 50% recombination frequency? (NEET 2013)
a) The genes may be on different chromosomes
b) **The genes are tightly linked**
c) The genes show independent assortment
d) If the genes are present on the same chromosome, they undergo more than one crossover in every meiosis.

32. Haploids are more suitable for mutation studies than the diploids. This is because (AIPMT 2008)
a) **All mutations, whether dominant or recessive are expressed in haploids**
b) Haploids are reproductively more stable than diploids
c) Mutagens penetrate in haploids more effectively than diploids
d) Haploids are more abundant in nature than diploids

33. Crossing over that results in genetic recombination in higher organisms occurs between (AIPMT 2004)
a) **Non-sister chromatids of a bivalent**
b) Two daughter nuclei
c) Two different bivalents
d) Sister chromatids of bivalents

## UNIT VIII – Biotechnology

1. What is the criterion for DNA fragments movement on agarose gel during gel electrophoresis? (NEET 2017)
a) **The smaller the fragment size, the farther it moves.**
b) Positively charged fragments move to farther end.
c) Negatively charged fragments do not move.
d) The larger the fragment size, the farther it moves.



TN_GOVT_BIOLOGY-BOTANY_XII_PAGES_200-238 CH 10.indd 213 3/14/2020 12:18:02 PM